xoji Quickstart Guide

5-Minute Setup for Xojo Developers

Slide 1: What is xoji?

AI-Friendly Project Indexer for Xojo

- Pre-indexes your Xojo project (100+ files)
- Reduces AI agent token usage by **5–8×**
- Enables agents to read only what's needed
- Automatic freshness checking
- Works with Claude Code, custom agents, build pipelines

Result: From 7% productive tokens → 50–60% productive tokens

Slide 2: The Problem xoji Solves

Without xoji:

```
AI Agent tasked with: "Fix crash in OrderForm"
↓
Scans 40+ class files blindly (~15% tokens)
↓
Reads entire 3,000-line window file (~35% tokens)
↓
Maps dependencies by brute force (~15% tokens)
↓
Finally writes code (~7% tokens) ← Only this matters!
```

90% wasted on overhead

Slide 3: The Solution

With xoji:

```
AI Agent tasked with: "Fix crash in OrderForm"
↓
Query codetree.json → "ListBox1.Change at line 112" (instant)
↓
Query dependencies.json → "OrderForm depends on: BaseForm, DatabaseManager"
↓
Read lines 100–140 only (~1 KB instead of 50 KB)
↓
Writes code
```

Agent gets exactly what it needs. 5–8× fewer tokens.

Slide 4: Installation (30 seconds)

macOS / Linux

```
git clone git@github.com:Jedt3D/xoji-project-parser.git
cd xoji-project-parser
./build.sh
# Binary ready: ./xoji
```

Windows

```
git clone git@github.com:Jedt3D/xoji-project-parser.git
cd xoji-project-parser
build.bat
REM Binary ready: xoji.exe
```

Slide 5: One-Command Project Setup

Automatically configure your Xojo project

```
xoji setup ../my_xojo_project
```

This does:

- ✓ Creates `.xojo_index/` directory
- ✓ Updates project's CLAUDE.md with instructions
- ✓ Installs `.claude/hooks/pre-task.sh` for auto-freshness checking

No manual editing required.

Slide 6: Build the Indexes

Create the JSON index files

```
xoji index ../my_xojo_project
```

Output:

```
.xojo_index/
├── codetree.json       ← Every method, property, line number
├── manifest.json       ← All files and their types
├── dependencies.json   ← Class relationships
└── meta.json          ← Project hash & timestamps
```

Time: ~200ms for 40+ files

Slide 7: Automatic Freshness (Optional)

Let Claude Code keep indexes fresh

The `.claude/hooks/pre-task.sh` hook (installed by `xoji setup`) runs automatically:

```
# Runs before each Claude Code task:
xoji check || xoji index

# If files changed: rebuild (silent)
# If fresh: do nothing
```

Zero manual intervention needed after setup.

Slide 8: Using the Indexes (For AI Agents)

Your project's CLAUDE.md now tells agents how to use xoji

Example:

```
# Instead of reading entire 3000-line window file:
cat .xojo_index/codetree.json | grep -A 5 "MainWindow"

# Get back:
# "Button1.Pressed": 78  ← line number

# Then read just what's needed:
sed -n '70,90p' AppSrc/MainWindow.xojo_window
```

Agents do this automatically when CLAUDE.md has xoji instructions.

Slide 9: Verify It Works

Test on your project

```
# Start fresh
xoji setup .
xoji index

# Check freshness
xoji check && echo "Fresh!" || echo "Stale"

# Inspect the indexes
cat .xojo_index/codetree.json | grep -o '"entity":"[^"]*"' | head -10
```

Slide 10: What Gets Indexed?

xoji extracts:

From .xojo_code (classes/modules):

- Methods (with line numbers)
- Properties (with line numbers)
- Events
- Constants
- Enums
- Inheritance & interfaces

From .xojo_window (UI files):

- All controls (buttons, textfields, listboxes, etc.)
- Window-level methods & properties

Slide 11: Example: Look Up a Method

Find `Button1.Pressed` in MainTestWindow.xojo_window

Step 1: Query the index

```
cat .xojo_index/codetree.json | grep -A 20 MainTestWindow.xojo_window
```

Result:

```
"events": {
  "Opening": 45,
  "Button1.Pressed": 78,
  "ListBox1.Change": 112
}
```

Step 2: Read only that section

```
sed -n '70,90p' AppSrc/MainTestWindow.xojo_window
```

Slide 12: Commands Cheat Sheet

```
# Setup & build
xoji setup ../project          # One-time configuration
xoji index ../project          # Build all indexes
xoji index --file path/file    # Update just one file (fast)

# Verify
xoji check ../project          # Exit 0 = fresh, 1 = stale

# Watch mode (coming soon)
xoji serve ../project          # Auto-index on file changes
```

Slide 13: Performance Numbers

Task	Time
Full index (40+ files)	~200 ms
Incremental (one file)	<100 ms
Freshness check (cached)	~50 ms
Binary size	3.7 MB
Index files total	~150 KB

Fast enough for real-time use.

Slide 14: File Structure (After Setup)

```
my_xojo_project/
├── .xojo_project                ← Project file
├── .xojo_index/                 ← Generated by xoji
│   ├── codetree.json
│   ├── manifest.json
│   ├── dependencies.json
│   └── meta.json
├── .claude/hooks/               ← Setup by xoji setup
│   └── pre-task.sh
├── CLAUDE.md                    ← Updated by xoji setup
├── App.xojo_code
├── AppSrc/
│   ├── MainWindow.xojo_window
│   └── ...
└── ...
```

Slide 15: Next Steps

You're ready!

1. **Install xoji**: `./build.sh` (macOS/Linux) or `build.bat` (Windows)
2. **Setup your project**: `xoji setup ../my_project`
3. **Build indexes**: `xoji index ../my_project`
4. **Use with Claude Code**:
   - Claude Code will read `.xojo_index/` automatically
   - Agents will use indexes to reduce token usage by 5–8×
5. **Optional: Verify**:

```
xoji check && echo "Fresh!"
cat .xojo_index/codetree.json | jq . | less
```

Slide 16: Support & Resources

Learn More:

- **Full docs**: `docs/README.md`
- **Architecture**: `docs/ARCHITECTURE.md` (coming soon)
- **GitHub**: https://github.com/Jedt3D/xoji-project-parser
- **Issues**: https://github.com/Jedt3D/xoji-project-parser/issues

Contact:

- Email: support@example.com
- GitHub Issues for bugs/features

Slide 17: Key Takeaways

✅ **xoji reduces AI agent token costs by 5–8×**

✅ **One-command setup**: `xoji setup`

✅ **Automatic freshness checking**: Pre-task hook keeps indexes current

✅ **Agent-friendly**: Indexes tell Claude Code what to read

✅ **Open source**: 100% pure Go, no external dependencies

✅ **Production-ready**: Used daily on 40+ file projects

Slide 18: Ready to Go!

Start now:

```
# 1. Clone & build
git clone git@github.com:Jedt3D/xoji-project-parser.git
cd xoji-project-parser
./build.sh

# 2. Setup your project (one time)
./xoji setup ../my_xojo_project

# 3. Index it
./xoji index

# 4. Your AI agents now use indexes automatically
```

Questions? Open an issue on GitHub.

Happy indexing! 🚀

This guide can be converted to PDF/slideshow format with tools like Marp, Pandoc, or reveal.js

```
# Example: Convert to PDF with Marp
npm install -g @marp-team/marp-cli
marp QUICKSTART.md -o QUICKSTART.pdf
```